

Decrypt https Traffic with Wireshark

In this article, the author demystifies the business of encrypting and decrypting network traffic, and translates arcane terms such as https, ssh, sftp, etc.



Wireshark, an interesting open source network sniffer, can not only read network traffic, but can further decrypt https traffic provided you have the private key! Let's learn more about decrypting https traffic using this tool.

To keep a network traffic sniffer from revealing login credentials, secure protocols such as https, ssh, sftp, etc, are used instead of their respective clear text protocols like http, telnet and ftp. Secure protocols encrypt traffic travelling between two end points; thus all the traffic available on the network is unreadable unless and until decrypted.

To understand secure protocols, let me try to explain the basics of encryption and decryption in simple terms.

Encryption and decryption

Encryption is a technique used to make readable (clear text) data unreadable (cipher text), typically using complex mathematical algorithms. Decryption is the exact reverse of the process of encryption.

The two main types of encryption

Symmetric key encryption: A single key is used to encrypt and to decrypt. Examples of the symmetric key algorithm

are the advance encryption standard (AES), data encryption standard (DES), etc. The typical length of a key is 128 bytes onwards, making encryption-decryption fast. However, the key must be known to both sides for successful operation.

Asymmetric key encryption: A key pair is used for encryption and decryption. Text encrypted by one key can be decrypted only by the second key and vice versa. Thus, the same key cannot perform encryption and its corresponding decryption. Examples of the asymmetric key encryption algorithm are Rivest, Shamir Adleman (RSA), ElGamal, Elliptic curve cryptography, etc.

Some important points about asymmetric key encryption are:

1. Asymmetric key encryption and decryption is slow compared to symmetric key encryption. For example, RSA encryption with a 1024-bit key is about 250 times slower compared to AES encryption with a 128-bit key.
2. One key (a private key) is kept confidential and the other key (the public one) is distributed.
3. Text encrypted by the sender's private key can be decrypted by the sender's public key. Thus, practically anybody can decrypt this since this key is distributed. However, please note that this ensures the authenticity of the sender—the private key is held confidential by